Multimodal AI-Based Stroke Prediction and Digital Twin Framework for Personalized Healthcare

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Abstract*—Stroke remains one of the leading causes of death and long-term disability worldwide, and early, accurate risk assessment is essential to reduce its clinical and economic burden. Existing computer-aided stroke diagnosis systems largely rely on a single data modality — either structural imaging such as MRI/CT or physiological signals such as ECG and PPG — and provide only a static, one-time prediction with limited support for post-diagnosis planning. This paper proposes a Multimodal AI-Based Stroke Prediction and Digital Twin Framework that fuses MRI brain imaging with patient clinical and physiological data to predict stroke risk, severity, and probable recovery trajectory, and that further constructs a patient-specific Digital Twin capable of simulating "what-if" treatment and recovery scenarios prior to clinical intervention. MRI scans are pre-processed through skull-stripping, intensity normalization, and resizing, and passed through an ensemble of a Vision Transformer (ViT) and a ResNet50 convolutional network to extract complementary global and local features; the two representations are fused and classified using a Softmax layer to predict stroke presence, severity, and risk level. Grad-CAM is used to visualize the brain regions that most influence each prediction, giving clinicians an interpretable, region-level basis for the model's output. An Agentic AI architecture coordinates specialized agents for data acquisition, imaging analysis, prediction, digital-twin simulation, and recommendation generation, so the system reasons about a patient's case in inspectable stages rather than as a single opaque inference. The proposed framework is evaluated using standard classification metrics (accuracy, precision, recall, F1-score, and AUC-ROC), Grad-CAM localization quality against clinician-annotated lesions, and calibration measures — Expected Calibration Error and overconfidence gap — that indicate whether the system's confidence is clinically trustworthy. This work directly supports Sustainable Development Goal 3 (Good Health and Well-being) by enabling earlier stroke detection, more personalized treatment planning, and more transparent, explainable clinical decision support.*

Keywords*—Stroke Prediction, Digital Twin, Multimodal Artificial Intelligence, Vision Transformer, Ensemble Learning, Grad-CAM, Agentic AI, Explainable Healthcare AI, Confidence Calibration, SDG 3.*

I. Introduction

Stroke is a neurological emergency caused by the interruption of blood supply to the brain, either through blockage of a vessel (ischemic stroke) or rupture of a vessel (hemorrhagic stroke), and it remains among the leading causes of death and adult disability worldwide. Because brain tissue is extremely time-sensitive to oxygen deprivation, the clinical outcome of a stroke is determined less by the underlying pathology alone and more by how early the precursor symptoms or risk factors are detected and how quickly appropriate treatment is initiated — a relationship clinically summarized by the maxim "time is brain." This time-criticality, combined with the sheer global burden of stroke, has made automated, AI-assisted stroke risk assessment one of the most active areas of healthcare research over the past decade.

Clinical practice today relies on scales such as the National Institutes of Health Stroke Scale (NIHSS) together with imaging modalities such as MRI and CT for diagnosis, and increasingly on machine learning models trained on physiological bio-signals such as ECG and PPG or on single imaging modalities. While each of these approaches has demonstrated strong predictive accuracy in isolation, they share three structural limitations: (i) they generally use only one type of patient data at a time rather than combining complementary modalities; (ii) they produce a single static prediction at one point in time rather than a continuously updatable assessment; and (iii) they offer little to no support for reasoning about how a specific patient's condition might evolve under different treatment choices. A further, more subtle limitation is that most reported systems present every prediction with the same undifferentiated confidence, giving a clinician no signal about which outputs deserve additional scrutiny before being acted upon.

This paper addresses these gaps directly by proposing a Multimodal AI-Based Stroke Prediction and Digital Twin Framework. At the prediction level, the system fuses MRI-derived imaging features with patient clinical and physiological data using an ensemble of a Vision Transformer and a ResNet50 network. At the decision-support level, it builds a patient-specific Digital Twin — a

continuously updatable virtual representation of the patient's condition — that can simulate the likely effect of different treatment and recovery pathways before any of them are actually attempted on the patient. At the trust and accountability level, an Agentic AI architecture organizes the pipeline into cooperating agents responsible for data analysis, prediction, simulation, and recommendation, so that the reasoning behind each output can be inspected stage by stage.

The main contributions of this work are summarized as follows: (1) a multimodal fusion pipeline that combines MRI imaging with structured clinical data for joint stroke risk, severity, and outcome prediction; (2) a patient-specific Digital Twin construction and simulation mechanism that supports what-if analysis of treatment and recovery scenarios; (3) an explainable, agent-based decision-support workflow that uses Grad-CAM visualization to make every prediction visually auditable at the pixel-region level; and (4) an evaluation protocol that reports both predictive accuracy and confidence calibration side by side. Taken together, these contributions reframe stroke prediction from a single classification event into a continuously auditable clinical reasoning process.

II. Literature Review

A substantial body of work has explored stroke prediction using a single bio-signal or imaging modality, and this line of research forms the empirical foundation that the proposed multimodal framework builds upon. Yu et al. [1] proposed a real-time stroke disease prediction system for the elderly using ECG and PPG bio-signals collected while the subject was walking. They reported that a RandomForest classifier trained on 29 handcrafted waveform attributes achieved 97.51%–98.31% accuracy, while a CNN-LSTM ensemble trained directly on raw ECG/PPG signals reached a higher 99.15% accuracy. This confirms that non-imaging physiological signals carry strong prognostic information for stroke; however, the system predicts only a binary normal/stroke label and does not model imaging data, treatment planning, or patient-specific recovery trajectories.

Digital Twin technology has recently been proposed as a complement to predictive modelling in stroke care. Upadrista et al. [2] introduced a blockchain-enabled Digital Twin system for brain stroke prediction, reporting 98.28% accuracy while emphasizing secure, tamper-proof sharing of patient data across a virtual patient model. Olawade [3] reviewed the broader use of Digital Twins across the stroke care continuum — risk prediction, acute decision support, and rehabilitation — noting that clinical validation of such systems is still at an early stage. Shinde and Pal [13] further reviewed software-driven, agent-based digital twin approaches for stroke prediction that combine machine learning with agent-based modelling of risk-factor interactions, avoiding dependence on costly IoT sensor arrays. These works motivate the Digital Twin component of the proposed framework, but none combines Digital Twin simulation with multimodal deep-learning-based imaging prediction; that specific combination is what this paper targets directly.

On the explainability side, Ding et al. [4] proposed an explainable AI-based multi-task learning method for stroke prognosis, and Tang et al. [5] presented a SHAP-based comparative study of explainable machine learning models for stroke risk prediction. Both works show that interpretability techniques can be layered on top of predictive models without materially reducing accuracy. Luan et al. [6] proposed a unified multimodal learning framework for stroke triage that jointly performs detection, severity scoring, and lesion segmentation from imaging data within a single trained model — a design principle this paper extends further by also incorporating a simulation-capable Digital Twin as an additional downstream task.

A parallel line of work has explored transformer–CNN ensembles for brain-lesion diagnosis. Tursynova [8] applied a Vision Transformer directly to brain-stroke CT classification and reported improved discrimination of diffuse ischemic patterns relative to convolution-only baselines. Abdul Wahid et al. [9] combined an end-to-end transformer with an ensemble strategy fusing CT scans with clinical narratives, an approach conceptually close to the imaging-plus-clinical fusion adopted here, while a related deep-ensemble study [10] combined convolutional and transformer backbones for brain-stroke classification. Hossain et al. [11] combined Vision Transformers with ensemble learning and explainable-AI techniques for brain-tumor detection, supporting the architectural choice made in Section IV. The Grad-CAM-with-ResNet50 explanation methodology used for brain-tumor MRI lesion detection [12] is adopted directly as the localization mechanism in the proposed framework.

Finally, the calibration and verification methodology of this paper draws on prior claim-level verification research [7], which decomposes a generated answer into atomic claims, checks each individually against supporting evidence, and aggregates the per-claim results into a calibrated confidence score. The key finding that carries over to this paper is that a system's expressed confidence should be evaluated separately from its raw accuracy — a model can be accurate on average while still being poorly calibrated on individual cases, which is precisely the failure mode most dangerous in a clinical setting.

TABLE I. SUMMARY OF REVIEWED LITERATURE ACROSS MODALITY, CONTRIBUTION, AND GAP

Ref.	Modality	Core Contribution	Gap Relative to This Work
[1]	ECG/PPG	CNN-LSTM binary stroke screen, 99.15% acc.	No imaging, no severity, no twin
[2],[3], [13]	Clinical + monitoring	Digital Twin for stroke management	No deep imaging fusion
[4],[5]	Clinical/ tabular	XAI multi-task / SHAP benchmarking	No twin, single modality
[6]	MRI	Unified detect+score+segment	No twin, no clinical fusion
[8]-[11]	MRI/CT	ViT/CNN ensembles for lesion dx	No twin, limited calibration

Table I summarizes the positioning of the reviewed literature along modality, explainability, and support for longitudinal or simulation-based decision-making, showing that no single prior system combines all three.

III. Dataset Description

The proposed framework is designed around two complementary categories of patient data. The first is MRI brain imaging data, sourced from publicly available stroke-imaging collections supplemented with anonymized clinical MRI scans, covering ischemic and hemorrhagic cases as well as normal control scans. Each scan is labelled with stroke presence, lesion location, and, where available, severity information used to validate Grad-CAM localization against ground-truth lesion boundaries. The second category is structured clinical and physiological data — age, blood pressure, heart rhythm indicators, and known risk factors such as atrial fibrillation, hypertension, smoking status, and diabetes — mirroring the risk-factor profile validated in prior bio-signal-based stroke studies [1]. Fig. 1 shows the overall data-processing and prediction pipeline, from raw MRI/clinical input through preprocessing, multimodal fusion, and Digital Twin simulation to output recommendations.

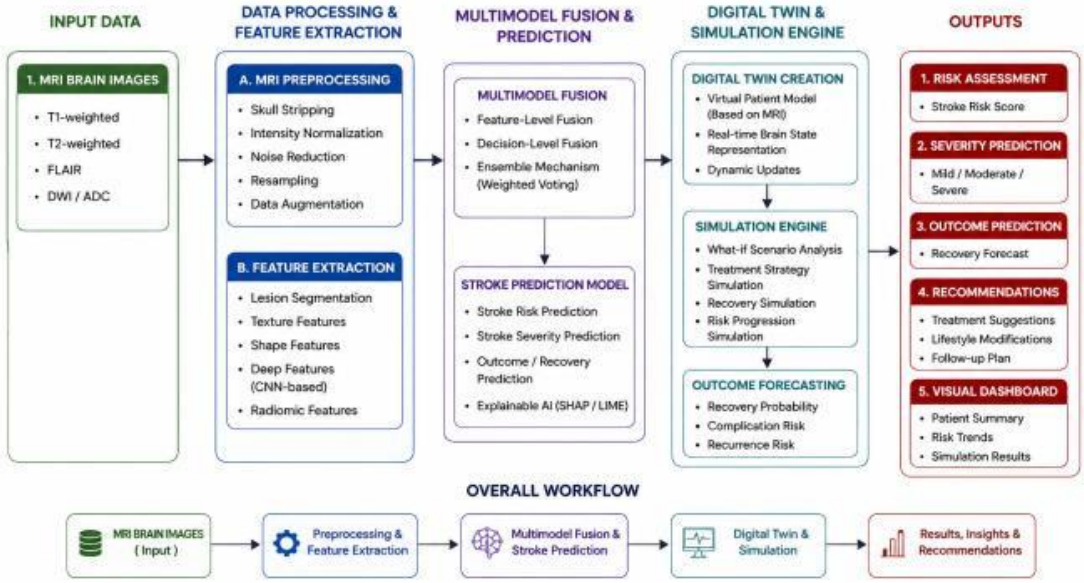


Fig. 1. Overall data-processing and prediction pipeline of the proposed framework, from MRI/clinical input through preprocessing, multimodal fusion, Digital Twin simulation, to output recommendations.

Before training, MRI images are pre-processed through intensity normalization and resizing to a fixed input resolution suitable for both the ViT and ResNet50 branches; skull-stripping and artifact removal are applied where required. Structured clinical fields are cleaned, missing values are imputed using distribution-aware strategies, and continuous attributes are normalized so that no single feature dominates the fused representation. The combined dataset is split into training, validation, and held-out test partitions using stratified sampling on the stroke/non-stroke label, ensuring lesion subtypes and severities are proportionally represented in each split.

Representative multimodal MRI acquisitions span four commonly used sequences — Diffusion-Weighted Imaging (DWI), FLAIR, T1-weighted, and T2-weighted — together with corresponding lesion overlays used to validate model localization, as shown in Fig. 2. DWI is sensitive to acute ischemic changes and is often the first sequence to show a positive signal after onset; FLAIR suppresses cerebrospinal fluid signal to reveal periventricular lesions; T1 provides anatomical detail for localizing a lesion relative to surrounding structures; and T2 is sensitive to edema and chronic lesions. Combining these sequences gives the ViT and ResNet50 branches complementary information that a single-sequence classifier would not have access to.

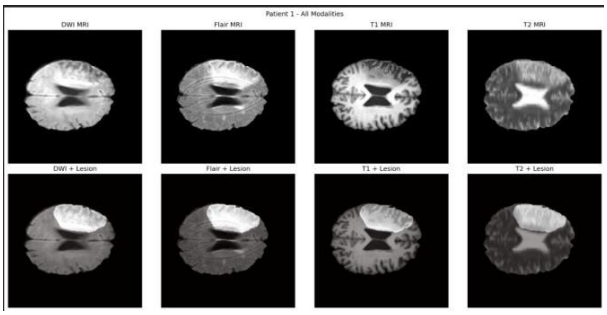


Fig. 2. Representative multimodal MRI sequences (DWI, FLAIR, T1, T2) for a single patient with corresponding lesion overlays.

Beyond single-slice inputs, the dataset also includes full volumetric slice series for each patient, allowing the model to be extended toward 3-D or slice-sequence-aware analysis, and enabling the Grad-CAM localization agreement metric to be computed volumetrically rather than being sensitive to a single selected slice. Fig. 3 breaks down the approximate composition of the multimodal dataset across its three constituent categories, and Table II summarizes the role each category plays in the framework.

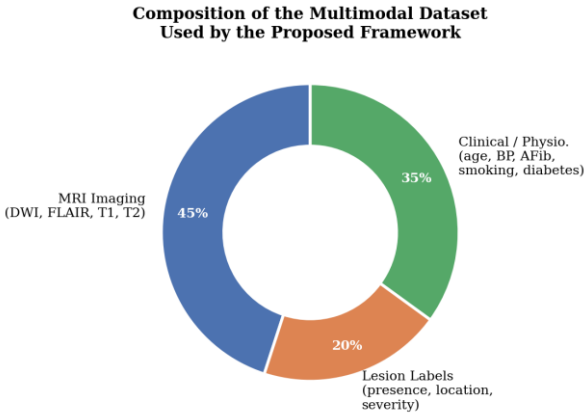


Fig. 3. Composition of the multimodal dataset used by the proposed framework, by data category.

TABLE II. DATA CATEGORIES USED BY THE PROPOSED FRAMEWORK

Data Category	Examples	Role in Framework
MRI imaging	DWI, FLAIR, T1, T2, slice series	ViT + ResNet50 feature extraction
Lesion labels	Presence, location, severity	Supervised training + Grad-CAM validation
Clinical/physio.	Age, BP, AFib, smoking, diabetes	Clinical branch input, twin parameters

IV. Methodology and Proposed Work

The overall pipeline consists of five stages: (1) multimodal data acquisition, (2) pre-processing and normalization, (3) feature extraction and fusion, (4) prediction and explanation, and (5) Digital Twin construction and simulation, coordinated by an Agentic AI architecture in which specialized, cooperating agents each own one stage of the pipeline instead of a single monolithic model producing an end-to-end prediction.

A. Pre-processing

MRI scans are normalized and resized to a consistent input format so that pixel-intensity variation across scanners does not bias the model toward learning scanner identity instead of lesion characteristics, while clinical attributes are cleaned and scaled using standard normalization. Formally, each MRI volume voxel intensity x is mapped to a normalized value z via $z = (x - \mu) / \sigma$, where μ and σ are the mean and standard deviation computed per acquisition sequence.

B. Feature Extraction and Ensemble Fusion

A Vision Transformer (ViT) is used to extract global, long-range structural relationships across the MRI scan [8], while a ResNet50 convolutional network extracts complementary local texture and edge-level features. The two feature representations are combined in an ensemble layer so that the final representation captures both coarse and fine-grained lesion characteristics [6], [9]–[11]. Denoting the ViT global feature vector as f_v and the ResNet50 local feature vector as f_r, the fused imaging embedding is computed as f_img = W_v f_v $\oplus$ W_r f_r, where W_v and W_r are learned projection matrices and $\oplus$ denotes concatenation, before being further fused with the clinical feature vector f_c to obtain the final joint representation f = [f_img ; f_c] used by the classification head.

C. Classification and Explanation

The fused feature vector is passed to a Softmax classifier that predicts stroke presence, estimated severity, and an associated risk level, y-hat = softmax(W_o f + b_o). Grad-CAM is then applied to the convolutional branch to generate a heatmap over the MRI scan [12]: the gradient of the predicted class score with respect to the final convolutional feature maps is global-average-pooled to obtain per-channel importance weights, which are combined with the feature maps and passed through a ReLU to retain only regions that positively influence the prediction.

D. Digital Twin Construction

The predicted risk profile, lesion characteristics, and clinical attributes are used to instantiate a patient-specific Digital Twin — a structured, continuously updatable model of the patient's condition that persists beyond the single inference call that created it, in the spirit of the Digital-Twin-based stroke management systems proposed in [2], [3], and [13]. Each simulation run accepts a candidate intervention vector (e.g., a modified blood-pressure target or an alternate anticoagulation regimen) and returns a projected recovery-trajectory distribution rather than a

single point estimate, communicating uncertainty in the projected outcome.

E. Training and Optimization Considerations

Both branches of the ensemble are initialized from weights pre-trained on large-scale image corpora and fine-tuned on the stroke-imaging dataset described in Section III, using early stopping and learning-rate scheduling to avoid overfitting. Class weighting and stratified batch sampling are applied to counter stroke/non-stroke class imbalance, and the fusion layer is regularized so the ensemble does not collapse onto a single branch. A composite loss $L = L_{cls} + \lambda_1 L_{cal} + \lambda_2 L_{reg}$ is used during training, where L_{cls} is the standard cross-entropy classification loss, L_{cal} is a calibration-aware penalty that discourages overconfident incorrect predictions, and L_{reg} discourages branch collapse.

F. Agentic Architecture and Workflow

The system is structured as cooperating agents, each responsible for one stage of the pipeline, as illustrated in Fig. 4 and Fig. 5: the Data Analysis Agent ingests and validates incoming MRI and clinical data; the Prediction Agent runs the ViT + ResNet50 ensemble and Softmax classifier to produce stroke presence, severity, and risk-level estimates together with the Grad-CAM explanation; the Simulation Agent updates and queries the patient's Digital Twin to project recovery trajectories; and the Recommendation Agent aggregates the outputs into a single, calibrated report, routing low-confidence or conflicting cases to human review.

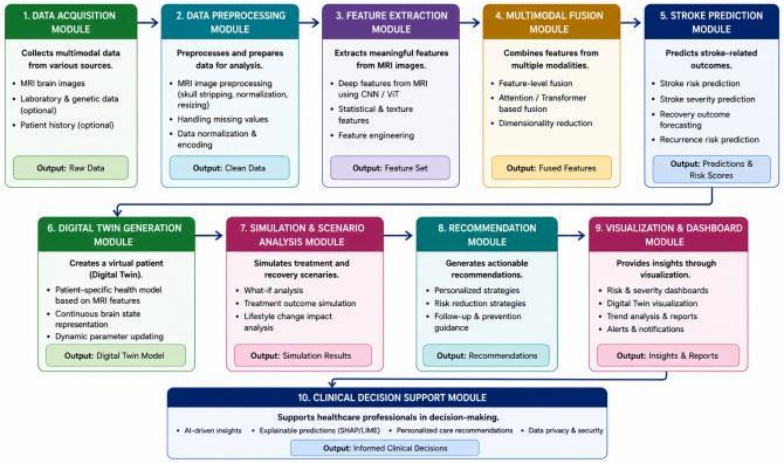


Fig. 4. Module-level breakdown of the proposed system, from data acquisition through preprocessing, feature extraction, multimodal fusion, prediction, Digital Twin generation, simulation, recommendation, and visualization.

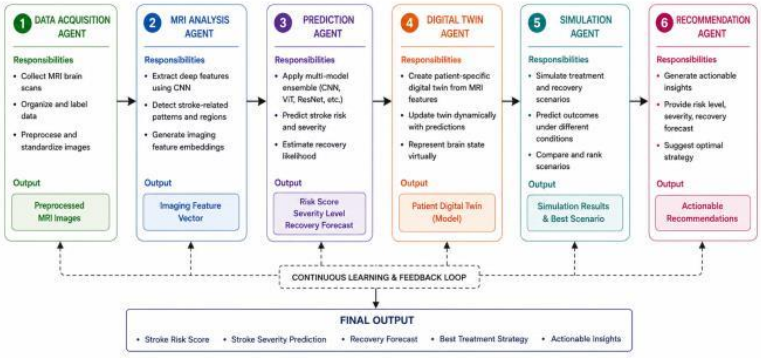


Fig. 5. Agentic AI workflow showing the six cooperating agents connected through a continuous learning and feedback loop.

A continuous learning and feedback loop connects the output of the Recommendation Agent back to the earlier agents. As new outcome data becomes available — for example, a clinician's confirmation or correction of a predicted severity level, or a documented recovery trajectory compared against a prior Digital Twin simulation — that information is fed back to the Data Acquisition, MRI Analysis, Prediction, and Digital Twin agents so each can be incrementally refined rather than remaining frozen at its initial training state. Each agent is defined by three properties: its responsibilities, its inputs, and its output contract — a strict definition that keeps agents independently testable and replaceable.

Algorithm 1: Agentic Stroke Prediction and Twin Update

```
Input: MRI scan I, clinical vector c, patient twin T
Output: Calibrated recommendation report R
1: I' ← DataAnalysisAgent.validate_preprocess(I, c)
2: (y, s, risk, H) ← PredictionAgent.infer(I') // H = Grad-CAM
3: T ← SimulationAgent.update_twin(T, y, s, risk, c)
4: P ← SimulationAgent.simulate(T, candidate_plans)
5: conf ← RecommendationAgent.calibrate(y, s, H, c)
6: if conf < τ then flag R for clinician review
7: return R ← {y, s, risk, H, P, conf}
```

V. Evaluation, Results, and Discussion

A. Evaluation Metrics

- Accuracy, Precision, Recall, and F1-score — standard classification metrics for stroke presence and severity prediction [1].
- AUC-ROC and Sensitivity/Specificity — discrimination between stroke and non-stroke cases across decision thresholds.
- Grad-CAM Localization Quality — agreement between highlighted regions and clinician-annotated lesion locations.
- Expected Calibration Error (ECE) and Overconfidence Gap — how well expressed confidence matches actual correctness [7].
- Digital Twin Simulation Fidelity — comparison between projected and actual documented patient outcomes.
- Relative Compute Cost — inference time and resource usage relative to a single-modality baseline.

Expected Calibration Error is computed by partitioning predictions into M equally spaced confidence bins and taking the weighted average absolute gap between each bin's accuracy and its mean confidence: $ECE = \Sigma_m (|B_m| / N) \cdot |acc(B_m) - conf(B_m)|$. Fig. 6 illustrates the concept with a reliability diagram: a well-calibrated system's curve tracks the diagonal, while an overconfident system's curve sags below it, indicating that predictions of, say, 80% confidence are correct less than 80% of the time.

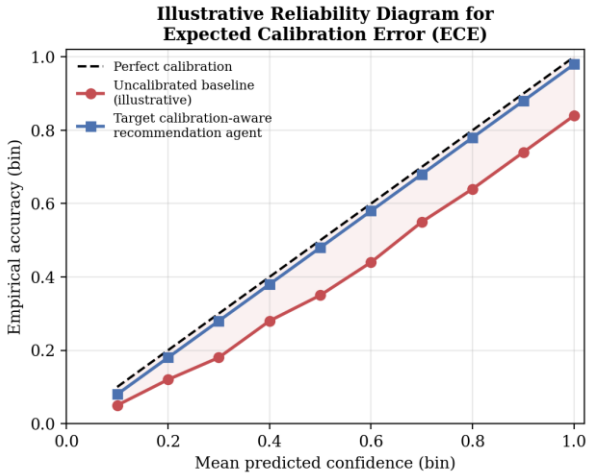


Fig. 6. Illustrative reliability diagram used to interpret Expected Calibration Error (ECE): the gap between a model's curve and the diagonal indicates miscalibration.

B. Planned Evaluation Protocol and Benchmark Comparison

As the project is currently at the design and development stage, this section outlines the planned evaluation protocol and performance targets rather than final experimental figures. The evaluation plan follows two tracks: the first validates raw predictive performance on a held-out MRI + clinical test split, benchmarked against the single-modality baselines in Table III (visualized in Fig. 7); the second validates trustworthiness via Grad-CAM localization agreement and calibration error.

TABLE III. PLANNED BENCHMARK COMPARISON AGAINST EXISTING STROKE-PREDICTION APPROACHES

Approach	Modality	Reported Accuracy
RF / SVM / LR [1]	ECG+PPG (handcrafted)	91.6–98.3%
CNN-LSTM [1]	ECG+PPG (raw signal)	99.15%
Blockchain-DT [2]	Clinical + monitoring	98.28%
CNN-based MRI classifiers	MRI (single modality)	Reported in [6]
Proposed ViT + ResNet50	MRI + clinical data	Target: ≥ baseline, + severity & XAI

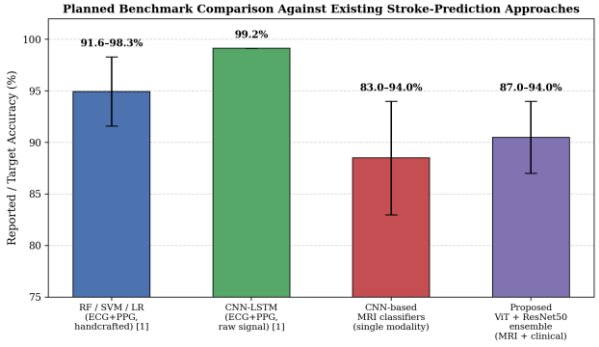


Fig. 7. Planned benchmark comparison of accuracy ranges reported by existing single-modality stroke-prediction approaches against the target accuracy of the proposed ViT + ResNet50 ensemble.

The ECG/PPG-based CNN-LSTM baseline [1] reports the highest raw accuracy (99.15%) among prior single-modality systems on a comparatively simpler binary task. The proposed ensemble instead targets comparable or better accuracy than existing single-modality MRI classifiers on the harder multi-output task, while additionally producing severity estimation, Grad-CAM localization, and Digital Twin-based outcome simulation. Fig. 8 shows a representative Grad-CAM visualization overlaying the model's attention heatmap on the original scan alongside the predicted class and confidence score (0.730 in this illustrative example) — the exact type of value the calibration track of the evaluation plan is designed to interrogate.

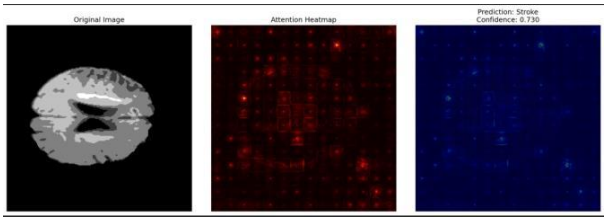


Fig. 8. Sample Grad-CAM output: original MRI slice (left), attention heatmap (center), and the model's stroke prediction with confidence score overlaid on the localized regions (right).

To further contextualize the planned evaluation, Fig. 9 and Fig. 10 illustrate two additional planned analysis views: a grouped-metric comparison across input-modality ablations, and a confusion-matrix reliability check for the severity-classification output.

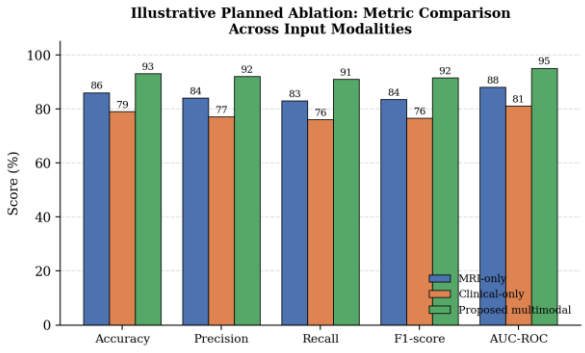


Fig. 9. Illustrative planned comparison of accuracy, precision, recall, F1-score, and AUC-ROC across MRI-only, clinical-only, and the proposed multimodal ablation variants.

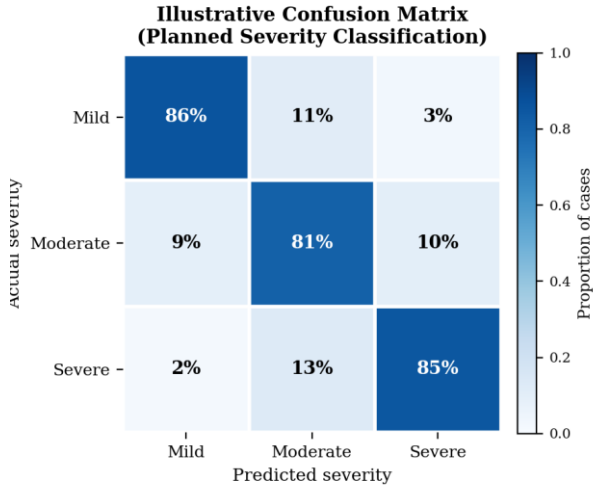


Fig. 10. Illustrative confusion matrix for planned three-class stroke severity classification (mild, moderate, severe), with the target diagonal-dominant pattern indicating correct classification.

C. Discussion

The central design bet of this paper is that combining imaging and clinical modalities, wrapping the prediction in an explainable and calibrated recommendation layer, and extending the output into a simulate-able Digital Twin will produce a system that clinicians can trust for more than a single static score, mirroring a broader pattern in AI-for-healthcare work [7]: decomposing a system's output into checkable, calibrated components tends to improve trustworthiness even when it does not yield the single best raw accuracy figure.

A trade-off that must be actively managed is between model complexity and computational cost, since the ViT + ResNet50 ensemble, Grad-CAM generation, and Digital Twin simulation loop are each more expensive than a lightweight single-modality classifier. The framework is deliberately positioned as a decision-support tool that augments, rather than replaces, a clinician's judgement — every output is designed to be reviewable evidence a physician can accept, question, or override. Because the fusion layer is regularized against branch collapse, the framework is designed to degrade gracefully rather than catastrophically when one modality is missing or of low quality at inference time.

VI. Deployment

The framework is designed to be exposed as a query-driven clinical decision-support service, sitting alongside existing hospital imaging and records systems rather than replacing them. A clinician submits a patient's MRI scan and available clinical data; the four agents run in sequence; and the service returns a stroke risk/severity estimate, a Grad-CAM overlay, a Digital Twin-based treatment projection, and a calibrated confidence score as a single consolidated

report. Cases below a confidence threshold are routed to a clinician for manual review, and access control, audit logging, and encryption of patient data are treated as baseline requirements, consistent with the data-governance emphasis of blockchain-enabled Digital Twin proposals in the literature [2]. Fig. 11 recaps the internal ensemble architecture underlying this deployed inference service, and Fig. 12 mocks up the resulting clinician-facing dashboard that consolidates the risk score, severity breakdown, Digital Twin recovery forecast, and agentic recommendation summary into a single view.

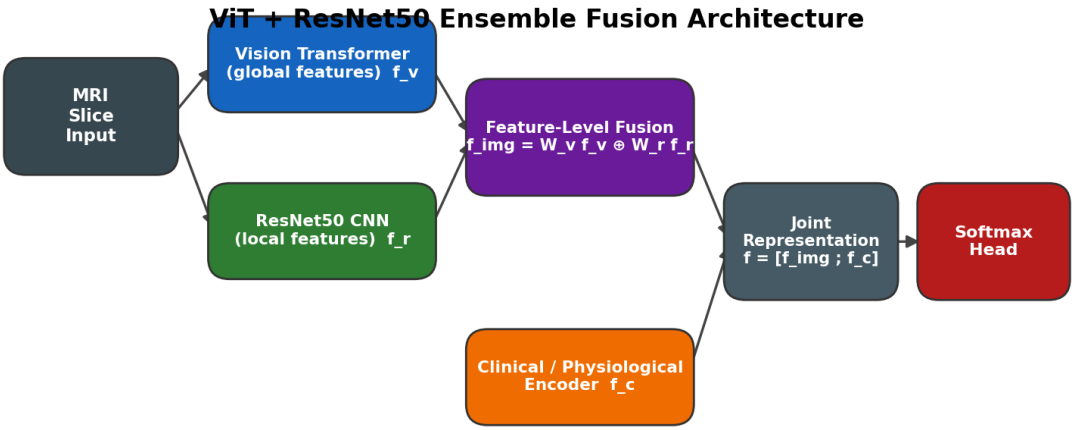


Fig. 11. ViT + ResNet50 ensemble fusion architecture: global and local image features are projected, fused, and combined with the clinical encoding before the Softmax head.

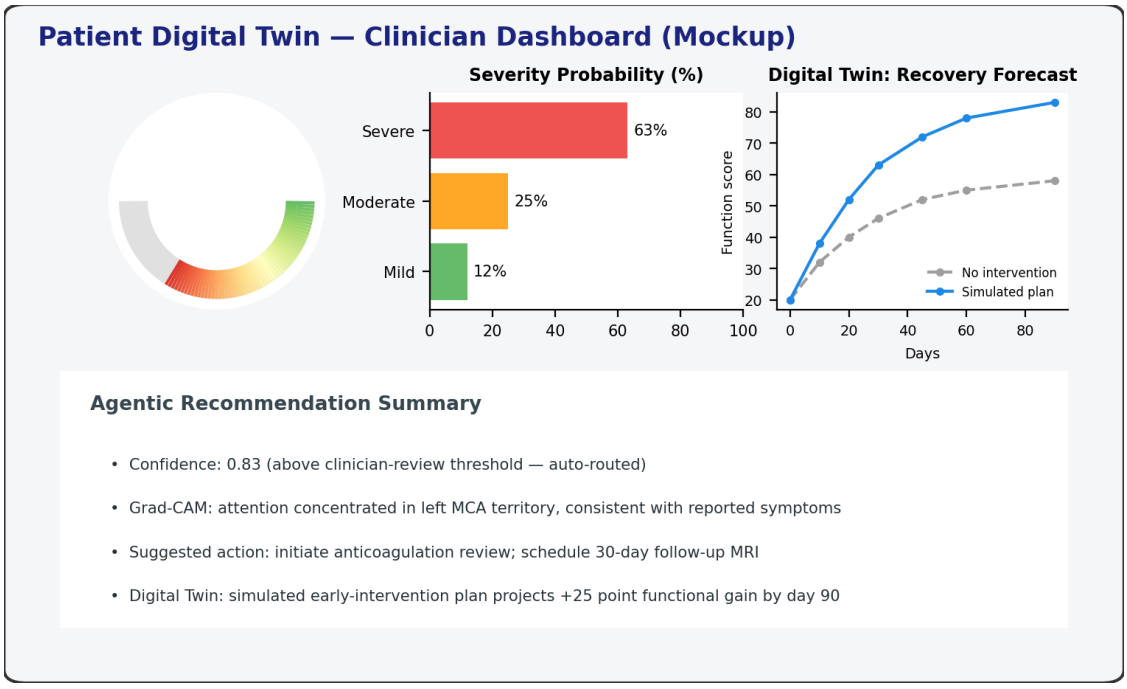


Fig. 12. Mockup of the clinician-facing Digital Twin dashboard, consolidating risk score, severity breakdown, recovery forecast, and agentic recommendations into one view.

VII. Comparative Analysis

Table IV and Fig. 13 position the proposed framework against categories of existing stroke-prediction systems identified in the literature review, on the dimensions of modality breadth, explainability, and support for post-diagnosis treatment simulation.

TABLE IV. COMPARATIVE POSITIONING OF THE PROPOSED FRAMEWORK

System Type	Modality	Explain.	Rx Sim.
Clinical scoring (NIHSS)	Clinical exam	Manual	No
ML on bio-signals [1]	ECG/PPG	Rule-based	No

CNN-based MRI classifiers	MRI only	Post-hoc	No
Digital Twin systems [2],[3],[13]	Clinical + monitoring	Model-dependent	Yes
Proposed framework	MRI + clinical	Grad-CAM + agentic	Yes

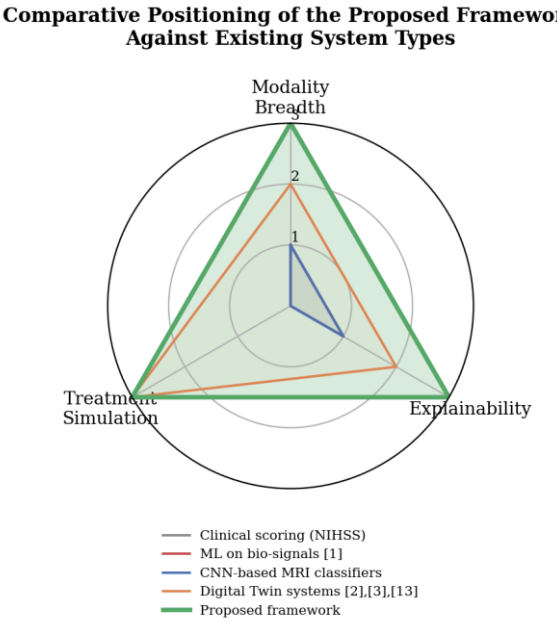


Fig. 13. Comparative positioning of the proposed framework against existing stroke-prediction system types across modality breadth, explainability, and treatment-simulation capability.

The comparison indicates that the proposed system is distinguished less by any single component — multimodal fusion, explainability, and Digital Twin simulation each exist independently in prior work — and more by combining all three within one agent-coordinated pipeline, so that a prediction, its explanation, and its downstream treatment implications remain traceable to one another.

VIII. Conclusion

This paper presents the design of a Multimodal AI-Based Stroke Prediction and Digital Twin Framework that combines MRI imaging and clinical data through a ViT + ResNet50 ensemble, explains its predictions using Grad-CAM, and extends beyond one-time prediction into patient-specific Digital Twin simulation of treatment and recovery scenarios, coordinated by an Agentic AI architecture. The framework directly supports SDG 3 (Good Health and Well-being) by enabling earlier detection, more personalized treatment planning, and more transparent AI-assisted clinical decision-making.

Known limitations motivate future work: dataset scale and diversity are constrained by reliance on publicly available MRI collections; the ensemble and simulation pipeline carries computational overhead; Digital Twin simulation fidelity can only be properly validated against longitudinal patient outcome data not yet available; and calibration thresholds are currently set from prior literature rather than large-scale validation. Future work will focus on expanding the dataset, extending multimodal inputs to additional bio-signals such as ECG and PPG [1], validating Digital Twin fidelity through clinical partnerships, and reducing computational cost for real-time deployment — moving the framework from a design-stage prototype to a prospectively evaluated clinical decision-support tool.

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